

Let X, Y be topological spaces. Two maps $f: X \rightarrow Y$ and $f': X \rightarrow Y$ are **homotopic** if one map can be continuously deformed to the other. (continuous)

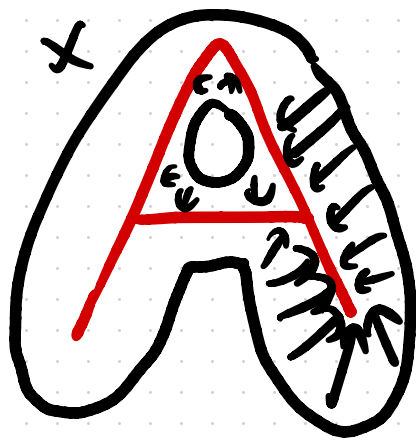
Formally, $f \simeq g$ if there exists a continuous function $F: X \times [0, 1] \rightarrow Y$ s.t. $F(x, 0) = f(x)$, $F(x, 1) = g(x) \forall x \in X$.

Two spaces X, Y are **homotopy equivalent** if there exist maps $f: X \rightarrow Y$, $g: Y \rightarrow X$ s.t. $f \circ g$ and $g \circ f$ are both homotopic to the identity map.

Let X be a topological space and A a subspace.

A **deformation retract** of X onto A is a map

$$F: X \times [0,1] \rightarrow X \text{ st. } F(x,0) = x, F(x,1) \in A \quad \forall x \in X$$
$$F(a,t) = a \quad \forall a \in A$$



X deformation retract onto A .

Prop: If X deformation retract onto A , then X and A are homotopy equivalent.

Theorem : If Δ and Δ' are simplicial complexes such that $|\Delta|$ and $|\Delta'|$ are homotopy equivalent, then $H_*(\Delta) \cong H_*(\Delta')$.

Simplicial
manifolds

Recall that a **manifold** of dimension d is a topological space where every point has a neighborhood homeomorphic to $\mathbb{R}^d$. A **simplicial manifold** is a simplicial complex homeomorphic to a manifold.

Manifolds: S^{n-1} spheres

$\mathbb{R}P^n$ real projective space

$\mathbb{C}P^n$ complex projective space

$S^1 \times S^1$ torus

Given a simplicial complex Δ and a face $\sigma \in \Delta$, define the **link** of σ to be

$$\text{link}_{\Delta}\sigma = \{ \tau \in \Delta : \tau \cup \sigma \in \Delta, \tau \cap \sigma = \emptyset \}$$

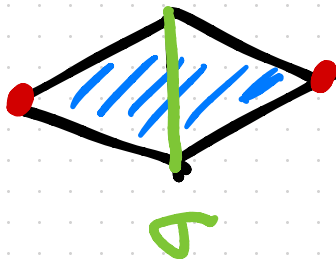
We define the **star** of σ to be

$$\text{star}_{\Delta}\sigma = \{ \tau \in \Delta : \tau \cup \sigma \in \Delta \}$$

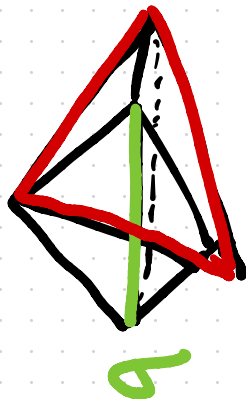
star
link



$$\text{link}_{\Delta}\sigma \approx S^1$$



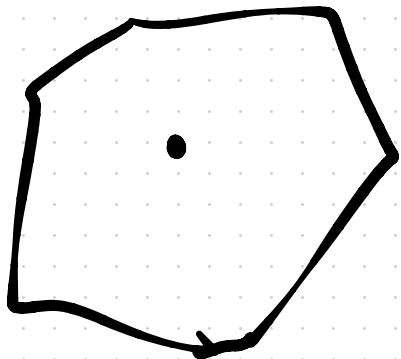
$$\text{link}_{\Delta}\sigma \approx S^0$$



$$\text{link}_{\Delta}\sigma \approx S^1$$

Proposition: If Δ is a simplicial d -manifold, then for all nonempty $\sigma \in \Delta$,

$$H_*(|K_\Delta \sigma|) \cong H_*(S^{d-1-\dim \sigma})$$

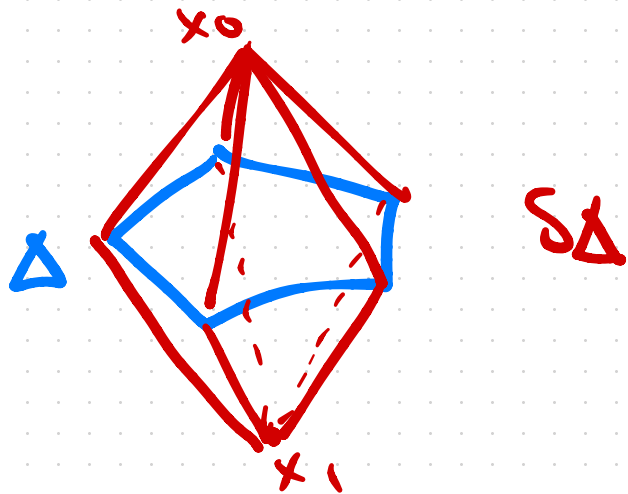


A space with the same homology as a sphere is called a **homology sphere**.

Not always true that $|K_\Delta \sigma| \approx S^{d-1-\dim \sigma}$
(when $\dim \geq 5$)

Given a simplicial complex Δ , the **suspension** of Δ is

$$S\Delta = \Delta \cup \{x_0 \cup \sigma : \sigma \in \Delta\} \cup \{x_1 \cup \sigma : \sigma \in \Delta\}$$



Theorem: If Δ is a homology sphere, then $S^2\Delta$ is homeomorphic to a sphere.

Let Δ be a homology 3-sphere which is not a sphere.

Then $S\Delta$ is a homology 4-sphere, not a sphere

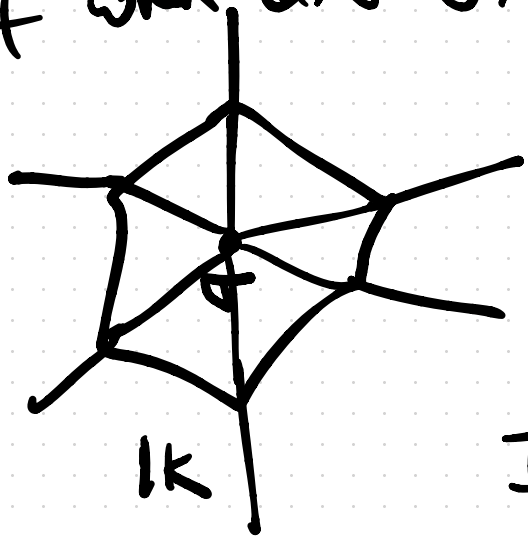
Then $S^2\Delta$ is a 5-sphere.

In $S^2\Delta$, the link of x_0 is $S\Delta$.

Proposition: If Δ is a simplicial d -manifold, then for all nonempty $\sigma \in \Delta$

$$H_*(IK_\Delta \sigma) \cong H_*(S^{d-1-\dim \sigma})$$

Proof when $\dim \sigma = 0$.



$$(IK_\Delta \sigma) \times [0, \infty) / IK_\Delta \sigma \times 0 \cong \mathbb{R}^d$$

If we puncture the origin,

$$(Ik_{\Delta^d}) \times (0, \infty) \simeq \mathbb{R}^d / 0$$

$$(Ik_{\Delta^d}) \times \mathbb{R} \simeq S^{d-1} \times \mathbb{R}$$

For any space X , $X \times \mathbb{R}$ is homotopy equivalent to X .

Thus Ik_{Δ^d} and S^{d-1} are homotopy equivalent.

Corollary: For Δ a simplicial manifold and all nonempty $\sigma \in \Delta$.

$$\sum_{i \geq -1} (-1)^i f_i(\text{lk}_{\Delta} \sigma) = (-1)^{d-1-\dim \sigma}$$

↑ reduced
Euler
characteristic
of $S^{d-1-\dim \sigma}$

$$\sum_{T \in \text{lk}_{\Delta} \sigma} (-1)^{\dim T} = (-1)^{d-1-\dim \sigma}$$

$$\sum_{\Delta \ni \sigma' \supset \sigma} (-1)^{\dim \sigma' - \dim \sigma} = (-1)^{d-1-\dim \sigma}$$

$$\sum_{\Delta \ni \sigma' \supset \sigma} (-1)^{\dim \sigma'} = (-1)^{d-1}$$